Green Purchase Intention Executive Summary

Predicting the Green Purchase Intention of Generation Z Café Consumers in Manila using Machine Learning Techniques

Prepared By	Kirk Nathaniel A. Gamorot
Date / Version	July 2026 | Version 1.0

Project Overview

This research investigates the factors influencing the green purchase intention of Generation Z café consumers in Manila using supervised machine learning techniques. By analyzing survey responses from 444 participants, the study compares six machine learning models to identify the most significant predictors of environmentally responsible purchasing behavior. The findings provide data-driven insights that can help cafés and sustainability-focused businesses better understand consumer preferences and develop more effective green marketing strategies.

Problem Statement

- Despite increasing environmental awareness among Generation Z, a significant gap remains between consumers' stated environmental intentions and their actual purchasing behavior, making it difficult for businesses to understand what truly influences green café patronage.
- There is limited local research in the Philippines that identifies the key factors driving Generation Z's green purchase intention using predictive machine learning techniques, particularly within the café industry.
- Without data-driven insights into consumer behavior, cafés and sustainability-focused businesses face challenges in developing effective marketing strategies and encouraging environmentally responsible purchasing decisions.

Research Objective

This study aims to identify and predict the key factors influencing the green purchase intention of Generation Z café consumers in Manila by applying and comparing six supervised machine learning algorithms. Through predictive analytics, the research seeks to determine the most influential behavioral, social, and environmental factors that drive sustainable purchasing decisions and provide actionable insights for businesses and researchers.

Dataset

- Target Population: Generation Z Café Consumers.
- Location: Manila, Philippines
- Respondents: 444
- Data Collection: Survey Questionnaire

Machine Learning Models

- Decision Tree
- Random Forest
- Gradient Boosting
- K-Nearest Neighbors
- Support Vector Machine
- XGBoost

Key Findings

- Random Forest and Support Vector Machine achieved the highest predictive performance, reaching an overall model accuracy of 81% in predicting Generation Z's green purchase intention.
- Subjective Norms, Green Perceived Quality, and Green Self-Identity emerged as the strongest predictors of green purchasing behavior, highlighting the importance of social influence, perceived product quality, and personal identity in sustainable consumer decisions.
- Environmental Awareness and Eco-Social Benefits were identified as the weakest predictors, suggesting that awareness alone is insufficient to influence purchasing decisions without stronger personal and social motivations.
- The study demonstrates that supervised machine learning provides an effective approach for analyzing consumer behavior and generating actionable insights that support sustainable marketing and business decision-making.

Business Impact

The findings provide cafés and sustainability-focused businesses with data-driven insights into the factors that influence Generation Z's green purchasing decisions. By identifying the strongest behavioral predictors, organizations can develop more effective marketing strategies, strengthen green branding initiatives, improve customer engagement, and support sustainable business practices through evidence-based decision-making.

My Contributions

- Proposed the research topic and objectives while contributing to the study's methodology and literature review.
- Assisted in developing and validating the survey instrument through questionnaire review, formatting, and pilot testing.
- Contributed to manuscript preparation by drafting portions of the Literature Review and Methodology, revising the paper, and performing quality assurance for formatting, citations, and overall consistency prior to publication.